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SECJ3553 – ARTIFICIAL INTELLIGENCE

SECTION 02

LECTURER: DR. ROLIANA BINTI IBRAHIM

NAME	MATRIC NUMBER
NURUL IKA SYAFINY BINTI AZHAR	A23CS0164
LUBNA AL HAANI BINTI RADZUAN	A23CS0107
NURAI SYAH BINTI MOHD ZIKRE	A23CS0160
NUR FIRZANA BINTI BADRUS HISHAM	A23CS0156
HARINI A/P SANGARAN	A23CS0081

Intelligent Agent

The intelligent agent used in MindConnect is a learning agent. The agent takes keystrokes as sensory input and acts on those inputs and displays output on the screen providing solutions to the users' problem.

PEAS Model Representation

This project is utilizing a learning agent type to control the system. Learning agents work to receive user input to understand users. Initially creating a psychological baseline for a user using a structured questionnaire. Subsequently, the agent will update the user's psychological condition using real-time engagement which is an AI-powered chatbot with the user to detect any sign of emotional trigger and determine the current state of the user's psychology. From there, the agent will produce an output as response based on the user's condition categorized from mild to critical to help the user in-need.

Table 1: PEAS Table Description

Performance	1. Accuracy of psychological trigger and emotion detection 2. Accuracy of psychological responses to the user
Environments	1. User reaction and response : User's emotional state and their subsequent behavior after receiving a reply
Actuators	1. Trigger Alert: Notifying the user of a detected pattern or reminder for the user's daily or weekly update. 2. Psychological Precautions: Delivering guides, reminders, or calming techniques to reduce the trigger impact. 3. Medical Assistance: Facilitating the user to connect with psychological experts.
Sensors	1. Keyboard input: User's response from questionnaire and chatbot. 2. NLP Analyzer: Understanding user's behaviour condition and pattern recognition. 3. System Clock: The time history when trigger commonly happen and its

	frequency.
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Diagram Representation

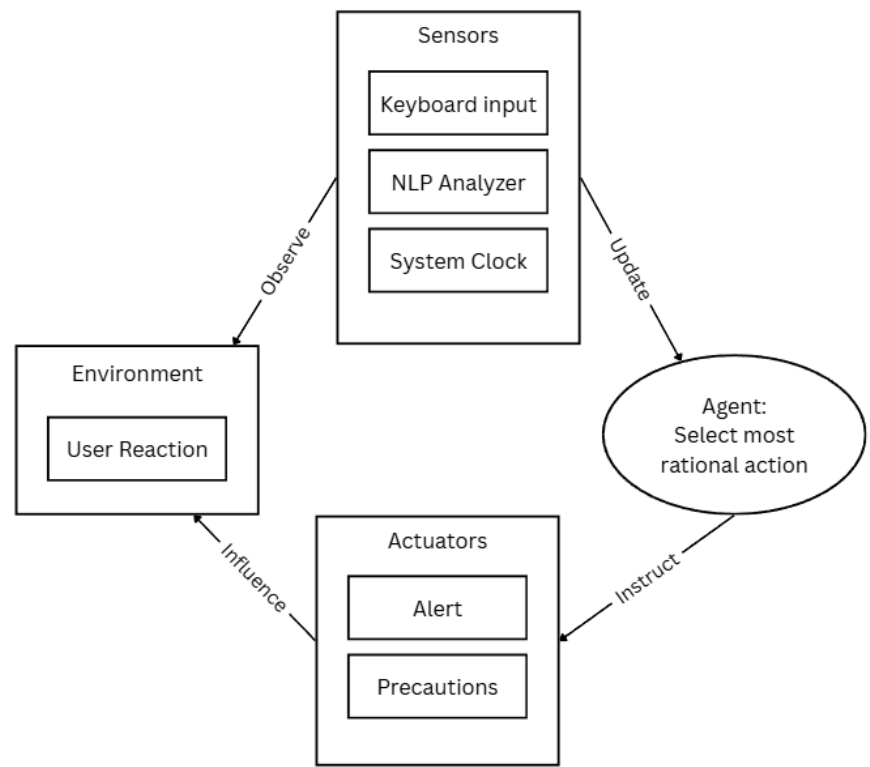


Figure 1: PEAS Diagram Representation

MindConnect is an AI application that detects and responds to user emotions based on the interaction in the application. Its performance is measured by how accurately it can identify the user’s emotional state from the input provided. Higher accuracy ensures the agent responds appropriately and effectively.

The process begins with MindConnect gathering information about users’ emotional state via questionnaire and chatbot. Thus, the environment will be a user reaction as it is being monitored. Keyboard input, Natural Language Processor (NLP) analyzer and system clock are the sensors of this application as it is the source of information from the users itself. Once MindConnect has

gained information via questionnaire and chatbot, the application categorises the severity of the condition and updates the agent. The agent of MindConnect then suggests appropriate ways of measure to be taken by users. For example, if the user's condition is mild then MindConnect provides suggestions, therapy and support via chatbot and if the condition is severe, then it suggests users to seek professional medical help. The suggestions and response given by the AI is considered as the actuators. After the conversation has ended with the users, MindConnect still monitors users' condition via daily check-ins and sending notifications if users are not responsive or not consistent, ensuring improving users' state of mind to the better.